Arista Access Node Hardware/Environmental Fault Resolution Procedure for Malleswaram Node (MALL-ACC-21)

Node: MALL-ACC-21 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_9340

Description

The Arista access node, situated at the Malleswaram Node (MALL-ACC-21), is susceptible to hardware/environmental faults, including ASIC memory parity errors, line card crashes, and power supply redundancy loss, triggering the ALARM_9340 warning. This failure domain is often precipitated by the intricate interplay between the node's hardware components and the environmental conditions in which it operates. Specifically, the Arista access node's reliance on ASICs for packet processing and its utilization of line cards for data transmission can lead to errors and crashes if these components are compromised. Furthermore, the loss of power supply redundancy can exacerbate these issues, resulting in node instability and potential downtime. The underlying protocol mechanics, such as the Spanning Tree Protocol (STP) and the Link Aggregation Control Protocol (LACP), can also contribute to the manifestation of these faults, as they rely on the proper functioning of the node's hardware components to maintain network stability and redundancy. Therefore, a comprehensive understanding of the node's architecture and the protocols it employs is essential for diagnosing and resolving these faults.

Pre-requisite Checks

- Verify the physical and environmental conditions at the Malleswaram Node, including temperature, humidity, and power supply stability.

- Check the status of redundant links and ensure that they are functioning correctly.

- Dispatch remote hands to the Malleswaram Node to perform on-site diagnostics and repairs if necessary.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces counters errors, Expected Output: Error
counters for each interface, Description: This command isolates errors
in the data plane by displaying error counters for each interface,
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```
allowing for the identification of interfaces experiencing errors and
the potential root cause of the fault.

Command: show processes cpu, Expected Output: CPU utilization for each
process, Description: This command provides insight into the control
plane by displaying CPU utilization for each process, enabling the
identification of processes consuming excessive CPU resources and
potentially contributing to the fault.

Command: show environment, Expected Output: Environmental conditions,
including temperature and power supply status, Description: This
command provides information on the environmental conditions at the
node, including temperature and power supply status, allowing for the
identification of potential environmental causes of the fault.
```

Resolution Steps

Step 1: Step: Replace faulty ASIC, Command: hardware module replace , Description: Replace the faulty ASIC with a new one to resolve ASIC memory parity errors. This step requires careful planning and execution to minimize downtime and ensure that the replacement ASIC is properly configured and integrated into the node.

Step 2: Step: Reset line card, Command: hardware module reset , Description: Reset the line card to resolve line card crashes. This step should be performed with caution, as it may cause temporary disruption to network services.

Step 3: Step: Restore power supply redundancy, Command: no power supply redundant, Description: Restore power supply redundancy to prevent future power supply redundancy loss. This step involves configuring the power supply settings to ensure that the node has a redundant power supply, minimizing the risk of downtime due to power supply failures.

Post-Check Verification

```
Command: show interfaces counters errors, Expected Output: Error
counters for each interface, Description: Verify that error counters
have decreased or stabilized after replacing the faulty ASIC or
resetting the line card.

Command: show processes cpu, Expected Output: CPU utilization for each
process, Description: Verify that CPU utilization has returned to
normal after resolving the fault.

Command: show environment, Expected Output: Environmental conditions,
including temperature and power supply status, Description: Verify
that environmental conditions, including temperature and power supply
status, are within normal ranges after resolving the fault.
```

Roll-back Procedures

- Step: Revert to previous ASIC configuration, Command: hardware module replace , Description: Revert to the previous ASIC configuration if the replacement ASIC causes issues or does not resolve the fault. This step requires careful planning and execution to minimize downtime and ensure that the previous ASIC is properly configured and integrated into the node.

- Step: Revert to previous line card configuration, Command: hardware module reset , Description: Revert to the previous line card configuration if the reset line card causes issues or does not resolve the fault. This step should be performed with caution, as it may cause temporary disruption to network services.

- Step: Revert to previous power supply configuration, Command: power supply redundant, Description: Revert to the previous power supply configuration if the restored power supply redundancy causes

issues or does not resolve the fault. This step involves configuring the power supply settings to ensure that the node has a redundant power supply, minimizing the risk of downtime due to power supply failures.